

Assignment 3

Date : _____

Posterior distribution of normal distribution with normal prior

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta, \sigma^2)$$

prior: $\theta \sim N(\mu, \tau^2)$

$$f(x_i | \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x_i - \theta)^2}{2\sigma^2} \right\}$$

$$L(\underline{x} | \theta) = (2\pi)^{-\frac{n}{2}} \sigma^{-n} \exp \left\{ -\frac{\sum_{i=1}^n (x_i - \theta)^2}{2\sigma^2} \right\} \quad \text{where } \underline{x} = (x_1, \dots, x_n)$$

$$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} \exp \left\{ -\frac{\sum_{i=1}^n x_i^2 + 2\theta \sum_{i=1}^n x_i - n\theta^2}{2\sigma^2} \right\}$$

$$\pi(\theta | \underline{x}) = \frac{L(\underline{x} | \theta) \pi(\theta)}{m(\underline{x})}$$

where $m(\underline{x}) = \int_{-\infty}^{\infty} L(\underline{x} | \theta) \pi(\theta) d\theta$

$$\pi(\theta) = \frac{1}{\sqrt{2\pi}\tau} \exp \left\{ -\frac{(\theta - \mu)^2}{2\tau^2} \right\}$$

~~$$m(\underline{x}) = \frac{1}{\sqrt{2\pi}\tau}$$~~

$$m(\underline{x}) = \int_{-\infty}^{\infty} (2\pi)^{-\frac{n}{2}} \sigma^{-n} (2\pi)^{-\frac{1}{2}} \tau^{-1} \exp \left\{ -\frac{\sum_{i=1}^n x_i^2}{2\sigma^2} - \frac{2\theta \sum_{i=1}^n x_i}{\sigma^2} - \frac{n\theta^2}{2\tau^2} \right\}$$

$$\exp \left\{ -\frac{\theta^2}{2\tau^2} + \frac{2\theta \mu}{\tau^2} - \frac{\mu^2}{2\tau^2} \right\} d\theta$$

$$= C \int_{-\infty}^{\infty} \exp \left\{ \theta \left(\frac{\sum_{i=1}^n \eta_i}{\sigma^2} + \frac{\mu}{\tau^2} \right) - \theta^2 \left(\frac{n}{2\sigma^2} + \frac{1}{2\tau^2} \right) \right\} d\theta$$

where $C = (2\pi)^{-\frac{(n+1)}{2}} \sigma^{-n} \tau^{-1} \exp \left\{ -\frac{\sum_{i=1}^n \eta_i^2}{2\sigma^2} - \frac{\mu^2}{2\tau^2} \right\}$

Let $2ab \frac{\sum_{i=1}^n \eta_i}{\sigma^2} + \frac{\mu}{\tau^2}$ & $b^2 = \frac{n}{2\sigma^2} + \frac{1}{2\tau^2}$

~~$$= C \int_{-\infty}^{\infty} \exp \left\{ a\theta - b^2\theta^2 \right\} d\theta$$~~

$$= C \int_{-\infty}^{\infty} \exp \left\{ -b^2\theta^2 + 2ab\theta - a^2 \right\} \exp \{a^2\} d\theta$$

$$= C \int_{-\infty}^{\infty} \exp \left\{ -(b\theta - a)^2 \right\} \exp \{a^2\} d\theta$$

Let $t = b\theta - a$
 $dt = b d\theta$

$$= \frac{C \exp \{a^2\}}{b} \int_{-\infty}^{\infty} \exp \{-t^2\} dt$$

$$m(\underline{\eta}) = \frac{C \exp \{a^2\}}{b} \sqrt{\pi}$$

$$\pi(\theta | \underline{\eta}) = \frac{C \exp \{a^2\} \exp \left\{ -(b\theta - a)^2 \right\}}{\frac{C \exp \{a^2\}}{b} \sqrt{\pi}}$$

$$= \frac{1}{\sqrt{2\pi} \cdot \frac{1}{\sqrt{2}b}} \exp \left\{ -\frac{\left(\theta - \frac{a}{b} \right)^2}{2 \left(\frac{1}{\sqrt{2}b} \right)^2} \right\}$$

$$\therefore \mathcal{O}_{\text{posterior}} \sim N\left(\frac{a}{b}, \frac{1}{2b^2}\right)$$

$$\frac{a}{b} = \frac{1}{2} \cdot \frac{2ab}{b^2}$$

$$= \frac{1}{2} \frac{\sum_{i=1}^n x_i + \frac{\mu}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

$$= \frac{\frac{n\bar{x} + \mu}{\sigma^2 + \frac{1}{\tau^2}}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

$$\frac{1}{2b^2} = \frac{1}{\frac{n^2}{\sigma^2} + \frac{1}{\tau^2}}$$

$$\therefore \mathcal{O}_{\text{posterior}} \sim N\left(\frac{\frac{n\bar{x} + \mu}{\sigma^2 + \frac{1}{\tau^2}}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}, \frac{1}{\frac{n^2}{\sigma^2} + \frac{1}{\tau^2}}\right)$$